

PRANAV UNNIKRISHNAN

PhD candidate

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[More about me](#)

Research interests

I study how evolutionary and ecological processes shape genomic diversity through time, integrating phylogenetic and population-genomic data with computational and statistical modelling to understand selection, adaptation, and diversification across species.

First author publications

1) Title:- 6Pgdh polymorphism in wild bulb mite populations: prevalence, environmental correlates and life history trade-offs

Journal name:- Experimental and Applied Acarology

Year:- 2024

Author list:- [Pranav Unnikrishnan](#), Szymon Grzesik, Magdalena Trojańska, Beata Klimek & Agata Plesnar-Bielak
<https://doi.org/10.1007/s10493-024-00909-4>

2) Title:- The interplay of environmental and social factors influences balancing selection: 6Pgdh in bulb mites

Journal name :- Currently in review (Ecology and Evolution)

Year:- In review

Author list:- [Pranav Unnikrishnan](#), Anna Spaeth, Magdalena Trojańska, Wiesław Babik, Agata Plesnar-Bielak
<https://doi.org/10.21203/rs.3.rs-4835969/v1>

Education

PHD IN BIOLOGY (2021 - 2025)

Jagiellonian University, Krakow, Poland

Doctoral Thesis: "Testing for factors maintaining 6Pgdh polymorphism in bulb mites"

Supervisor: Prof hab. Dr. Wiesław Babik

Assistant supervisor: Dr. Agata Plesnar-Bielak

INTEGRATED BS-MS (2015 - 2020)

Indian Institute of Science Education and Research, Tirupati, India

Masters Thesis: "Evolution of coat color and pattern in squirrels of the world"

Supervisor: Prof. Dr. Nandini Rajamani

SECONDARY EDUCATION

BSS GHSS, Palakkad, Kerala, India

Research experience

PHD

Title:- Maintenance of 6Pgdh Polymorphism in Bulb Mites: Insights from Field Studies, Experimental Evolution, and Life-History Trade-Offs

Keywords:- Sexual selection, experimental evolution, balancing selection

Study system:- Bulb mites (*Rhizoglyphus robini*)

- Investigated how **environment and sexual selection** interact to **maintain polymorphism** at the 6Pgdh locus in *Rhizoglyphus robini*.
- Combined **multi-year field surveys, experimental evolution, and life-history assays** to test the maintenance of 6Pgdh polymorphism.
- Modelled allele-frequency dynamics across **temperature and sex-ratio treatments**, showing a significant temperature × sexual selection interaction consistent with **environment-dependent balancing selection**.
- Developed reproducible population-genomic workflows for **SNP genotyping, allele-frequency estimation, and selection inference** (PoPoolation2, R).
- Conducted **RNA-seq analyses** comparing 6Pgdh genotypes, including quality control (FastQC, Trimmomatic), read mapping (STAR), variant calling, differential expression (DESeq2), and GO enrichment (goseq).
- Identified genotype-specific expression modules linked to **stress response and reproduction**, integrating transcriptomic and population-level patterns to build a mechanistic view of balancing selection.
- Detected a **temperature × sexual selection interaction driving allele-frequency dynamics**, confirming **balancing selection under variable environments**.

MASTER'S THESIS

Title:- Explaining the evolution of color and patterns in squirrels using meta-analyses and experimental approaches

Keywords:- Coat color evolution, phylogenetics, meta-analysis

Study system:- Family *Sciuridae* (squirrels)

- Compiled a comprehensive database of **280+ squirrel species**, integrating morphological, ecological, and climatic data for **comparative analyses**.
- Constructed **species- and genus-level phylogenetic trees** using concatenated mitochondrial and nuclear genes (cytb, 12S, 16S, COX1) with partition-specific substitution models and ultrafast bootstraps (IQ-TREE, ModelFinder).
- Performed **ancestral state reconstruction and stochastic mapping** to infer the evolutionary history of coat colour and pattern; tested for **phylogenetic signal** (phytools, fitDiscrete).
- Applied phylogenetic ANOVA and correlation analyses to test **Gloger's rule**, linking **colour evolution to environmental gradients** such as precipitation and evapotranspiration.
- Combined large-scale comparative datasets with computational and experimental approaches to **model trait evolution across lineages**.
- Found that **darker coat colours were associated with higher precipitation and evapotranspiration**, supporting Gloger's rule; patterning conferred camouflage advantages in visually complex environments

- # Awards & Scholarships

2023	Visibility and Mobility module (travel grant) to attend SMBE 2023
2022	Outstanding student poster in ESEB 2022
2020	NCN Scholarship for three years
2020	All India Rank of 52 in GATE (Graduate Aptitude Test in Engineering) in Ecology and Evolution
2015	DST Inspire Scholarship for five years

Major Presentations & conferences

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|-----------|-------------------------------|
| PEC 2023 | :- Oral + Poster presentation |
| EMPSEB 28 | :- Oral + Poster presentation |
| SMBE 23 | :- Poster presentation |
| PEC 2022 | :- Poster presentation |
| ESEB 2022 | :- Poster presentation |

Skills

- **Integrative evolutionary research:** Experienced in linking field observations, experimental evolution, and genomic data to study how selection and environment shape genetic diversity.
- **Comparative and phylogenetic analysis:** Skilled in using evolutionary frameworks to infer trait and genotype evolution across species and timescales.
- **Data-driven evolutionary inference:** Experienced in integrating genomic, transcriptomic, and environmental datasets using computational models to identify evolutionary patterns.
- **Computational and quantitative reasoning:** Proficient in developing reproducible analytical workflows, applying statistical models, and interpreting large-scale population and transcriptomic datasets.
- **Experimental design and interpretation:** Designed and conducted multi-year selection experiments to quantify genotype-by-environment interactions
- **Research communication and collaboration:** Regularly present work at international conferences; experienced in writing manuscripts and coordinating collaborative projects.
- **Project management:** Capable of planning and executing multi-stage research projects, managing datasets, timelines, and resources effectively.

Software and languages:-

- R, Python (HPC)
- FastQC, Trimmomatic, STAR, PoPoolation2
- QGIS
- IQ-TREE, MEGA X, Geneious
- ImageJ, RAVEN, BORIS

Lab techniques:-

- DNA & RNA extraction
- PCR, RT-PCR, gel electrophoresis
- RNA-seq library preparation
- Animal handling & tagging

INTERNSHIP

- Title:-** Effects of stress (chemical and physical) on the migration ability of experimentally evolved fruit flies
- Keywords:-** stress response, behavioral ecology, life history traits
- Study system:-** *Drosophila melanogaster*
- Applied experimental **stress treatments** to base and dispersal-selected *Drosophila* lines, targeting sleep disruption (via caffeine) and mechanical disturbance (via vibration).
 - Assessed **behavioral and reproductive traits post-stress**, including activity levels, development time, sperm competition, and copulation duration.
 - Evolved lines exhibited greater reproductive performance and activity under stress, suggesting **enhanced tolerance linked to dispersal adaptation**.

INTERNSHIP

- Title:-** Territory ranges of two newly discovered bird species
- Keywords:-** Territory mapping, ecological hotspot, bird watching
- Study system:-** Birds

- A **field based study** on Kodaikanal, an ecological hotspot in India rich in biodiversity.
- The study was on **two newly discovered bird species**, to map their territories and to find the extend of their range.
- The data obtained from the project helped **conservation efforts** and also in planning for further studies on the species.

Other activites

ORGANIZATIONAL ACTIVITIES

- 1) **Name:-** EMPSEB29 (29th edition of the European Meeting for PhD Students in Evolutionary Biology)
Role:- Organizer + Designer
- 2) **Name:-** PEC 2023 (Polish Evolutionary Conference)
Role:- Volunteer
- 3) Organizing activities for events such as Science day, mini workshops on biodiversity, R-workshop, in university

TEACHING EXPERIENCE

- 1) **Course name:-** Effective research communication
Institution:- Jagiellonian University
Role:- Tutor **Duration:-** 3 years (Summer semesters)
- 2) **Course name:-** Ecology and evolution (I and II)
Institution:- IISER, Tirupati
Role:- Teaching assistant **Duration:-** 1 year (full year)
- 3) Mentored and supervised multiple students involved in the project related to the PhD. Additionally, I have completed courses on effective teaching and student communication.

ARTISTIC AND PERSONAL PURSUITS

- 1) **Drawing & Illustration**
Pursue both traditional sketching and digital art as a personal hobby, a way to relax and express creativity.
- 2) **Dance**
Member of a dance school ensemble for 2+ years, performing on stage in group productions

References

Prof hab. Dr. Wiesław Babik:- wieslaw.babik@uj.edu.pl
Dr. Agata Plesnar-Bielak :- agata.plesnar@gmail.com